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AVOIDING REGULATION:
EVIDENCE FROM NURSING HOMES

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ABSTRACT

We study strategic responses by firms to threshold-based regulations. We document extensive bunching in response to additional staffing requirements for California nursing homes with at least 100 beds. To estimate the regulation's effect on staffing, we develop a novel treatment effects estimator for threshold-based regulations under precise manipulation, by decomposing firms into response types based on counterfactual and realized staffing and size, separating causal effects from selection. We estimate that higher staffing above the threshold is nearly entirely due to selection, with low-staffing facilities manipulating their size to avoid the regulation. These size distortions limit access, particularly for disadvantaged patients.

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1 Introduction

Threshold-based regulations are widely used across policy domains, yet their effectiveness depends critically on the extent to which entities can manipulate whether they are subject to the regulation. When treatment status is tied to whether an assignment variable crosses a discrete threshold—as it is for labor regulations (Schivardi and Torrini, 2008; Garicano et al., 2016; Caicedo et al., 2022), tax and benefit rules (Onji, 2009; Dillender et al., 2022), financial regulations (Iliev, 2010; Ewens et al., 2024), antitrust scrutiny (Wollmann, 2019), and many other settings—entities often have strong incentives to bunch just below the threshold. This avoidance behavior can undermine the effectiveness of the regulation, particularly because the entities that select into avoidance are often those that are furthest from compliance. Furthermore, this avoidance behavior may itself have important consequences. The welfare effects of a threshold-based regulation thus depend on two key objects: the effects of avoidance behavior by bunchers, as well as the effects of compliance behavior among entities that do not manipulate their assignment variable.

In this paper, we develop a simple framework to estimate the effects of a threshold-based staffing regulation in nursing homes, a sector with annual spending exceeding \$200 billion (Martin et al., 2025). We focus on a California regulation requiring that nursing homes with 100 or more beds maintain a registered nurse on-site 24 hours per day. Higher nursing home staffing is often linked to better patient health outcomes, but hiring more staff is costly to facilities. Because the staffing requirement we study only applies to facilities with 100 or more beds, facilities for which meeting the staffing requirement would be costly may choose to avoid the requirement by adjusting their scale and capping their bed count below 100.

Estimating the welfare effects of this threshold-based policy requires identifying the effects of both avoidance and compliance behaviors. First examining avoidance behavior, we document extensive bunching in facility size to avoid the regulation: facilities in California are 158 times as likely to have 99 beds as 100 beds, with no such pattern in other states. We also estimate that the probability of bunching is decreasing in the costs of manipulation, as bunchers tend to have both lower staffing and smaller size absent the regulation. On average, facilities bunching below the 100-bed threshold would have had 113.8 beds in the absence of the regulation. This shrinkage results in reduced patient access, particularly in low-income, high-Hispanic ZIP codes, suggesting that the welfare effects of avoidance behavior are negative.

Identifying the second object of interest—the effect of compliance with the staffing regulation—is more difficult because facilities’ ability to precisely manipulate their assignment variable invalidates standard regression discontinuity research designs (McCrary, 2008; Lee and Lemieux, 2010). The key econometric challenge is that staffing levels of facilities above the threshold may be high due to the treatment effect of the regulation on staffing or because only facilities that would already

have had high staffing without the regulation choose to remain above the threshold. In order to identify the effect of the regulation on staffing levels, we develop a novel method for estimating causal effects in the presence of selection in avoidance behavior that can be applied even beyond our setting.

Using a simple model of facility utility over size and staffing that formalizes the tradeoff between complying with staffing mandates and operating at one’s preferred scale, we partition facilities into four distinct response types: never-regulated, self-regulated, bunchers, and rule-takers. We show that under reasonable assumptions about continuity and smoothness of potential outcomes, we can characterize these response types, both in terms of baseline characteristics as well as counterfactual and realized values of staffing and size. The key insight for estimating the effect on staffing is that while staffing outcomes above the threshold reflect both selection and treatment, the degree to which bunching facilities differ from those below the manipulation region is completely attributable to selection. Thus, we compare staffing at facilities bunching below the threshold to those with slightly fewer beds, identifying the degree to which bunching facilities are selected on staffing levels. With the degree of selection in hand, the remaining difference in outcomes for facilities just above and below the regulatory threshold can be attributed to the treatment effect of the regulation. In short, we recover counterfactual outcomes for the facilities subject to the regulation net of selection using bunchers as a selection-revealing group.

We find that facilities bunching below the threshold have significantly lower staffing per bed-day than facilities just below the threshold that do not manipulate their size, employing 9.4% fewer RN hours per bed-day. Yet among the facilities that comply with the regulation, we detect little change in actual staffing, with nearly the entire observed difference in staffing being attributable to selection. This null effect on staffing, combined with the reductions in access from bunching, indicates that the policy we study reduces social welfare.

This paper contributes to two main strands of research. Methodologically, we contribute to the literature that uses policy thresholds for identification by bridging regression discontinuity and bunching designs. Bunching at a discontinuous threshold often undermines the ability to estimate the effect of crossing the threshold ([McCrary, 2008](#); [Imbens and Lemieux, 2008](#)), while this avoidance behavior is sometimes the phenomenon of interest ([Saez, 2010](#); [Chetty et al., 2011](#); [Kleven and Waseem, 2013](#); [Benzarti, 2020](#); [Goff, 2024](#)). By contrast, we estimate the effect of crossing the regulatory threshold even in the presence of bunching—which typically invalidates regression discontinuity designs—by identifying the degree of selection into bunching ([Caetano, 2015](#); [Diamond and Persson, 2016](#); [Caetano et al., 2025](#)). While existing research has shown that bounds on the treatment effect of the policy can be obtained under milder assumptions ([Gerard et al., 2020](#)), we use a simple model and somewhat stronger assumptions to generate transparent reduced-form tools for point identification of causal effects in the presence of bunching that can be

easily used in a variety of settings by applied researchers.

We also contribute to the literature on the effectiveness of nursing home staffing regulations. Prior research has focused on attempts to raise staffing through financial incentives such as direct payment or competition ([Hackmann, 2019](#); [Gandhi et al., 2024](#)), or through staffing mandates that apply to all facilities ([Park and Stearns, 2009](#); [Matsudaira, 2014](#); [Chen and Grabowski, 2015](#)), often finding mixed results. By contrast, we focus on a threshold-based regulation, finding that strategic responses by facilities almost fully undermine its intended effect on staffing. Moreover, we find an unintended effect of the regulation on a margin separate from staffing: reduced access for minority residents.

2 Setting and Data

2.1 Nursing Home Staffing

Nursing homes provide care for roughly 1.2 million Americans ([Chidambaram et al., 2025](#)), and more than half of individuals aged 57–61 today are expected to spend some time in a nursing home during their lives ([Hurd et al., 2017](#)). Given the vulnerable population served, policymakers have long been concerned with nursing home quality generally and registered nurse (RN) staffing in particular: RNs oversee clinical care and respond to emergent health issues, and have been shown to improve care quality ([Lin, 2014](#)) and patient survival ([Friedrich and Hackmann, 2021](#)). At the federal level, regulations require that all Medicare-certified nursing homes have an RN on duty for at least 8 hours per day,¹ with facilities with more than 120 beds being required to employ a full-time qualified social worker.²

Many states have adopted stricter staffing requirements, with California standing out for its relatively stringent rules. In addition to regulating the minimum number of care hours per patient for all facilities, California also sets higher requirements based on the size of the facility. In particular, while all California nursing homes are required to have an RN or licensed vocational nurse (LPN) on duty 24 hours a day, facilities with 60 or more beds are required to have a separate director of nursing services, while facilities with 100 or more beds are required to have 24-hour RN (rather than LPN) staffing.³ RN staffing is higher-skill than LPN, but also much more expensive. In our data, the average RN wage for a California facility is just over \$42 an hour versus only \$31 for LPNs. This means the annual cost of 24 hours of RN coverage is over \$100,000 more than the same amount of LPN coverage. Thus, the staffing rule potentially introduces a strong incentive for

¹See 42 C.F.R. §483.35. In 2024, the Biden administration promulgated a 24-hour RN staffing requirement, but this was repealed by the Trump administration.

²See 42 C.F.R. §483.70(o).

³See Cal. Code Regs. Tit. 22, §72329.

facilities to manipulate their scale to avoid being subject to the regulation.

Importantly, there are no other discontinuous regulations of California nursing homes at the 100-bed threshold. In addition to the staffing requirements discussed above, the only other size-based regulations imposed by California on nursing homes are at 6, 50, 60, and 200 beds.⁴ These regulations generally relate only to facilities with intermediate care units, or are easily met, such as the requirement to have a telephone. Thus, any discontinuity in staffing or bunching at 100 beds can be plausibly attributed to the 24-hour RN staffing requirement.

2.2 Data

We use three national, facility-level administrative data sources: the Healthcare Cost Report Information System (HCRIS), the Online Survey Certification and Reporting (OSCAR) system, and the Payroll-Based Journal (PBJ) staffing data. HCRIS is maintained by the Centers for Medicare & Medicaid Services (CMS) and includes cost reports submitted annually by all Medicare-certified facilities. These reports contain data on staff hours and wages and facility bed count.

The OSCAR system provides inspection and survey data collected by state licensing agencies. These data include facility characteristics such as for-profit status, chain ownership, and the presence of an Alzheimer's unit. They also allow us to validate bed counts and assess an alternative measure of staffing.

Finally, the PBJ staffing data, which are only available starting in 2017, offer further measures of staffing that are thought to be the most accurate of our three data sources (Geng et al., 2019). We merge these three data sources using facility identifiers, creating a panel that captures both facility size and staffing levels.

Our key measure of staffing is RN hours per bed-day. RN hours are the staffing level subject to the regulation, and scaling facility-wide reported staffing by bed-days allows for a measure of staffing that is not mechanically related to facility size. While the regulation requires 24-hour RN staffing, we observe only total RN hours per day. Thus, our outcome measures whether facilities increased RN staffing in response to the regulation and abstracts from any reallocation of RN hours within each day.

We report estimates using staffing data from each data source independently, with the PBJ data as our primary source. The assignment variable is number of beds, which we validate is consistently reported in both the HCRIS and OSCAR data.

We also use geographic information on facility locations combined with data on ZIP code-level demographics from the American Community Survey. These data allow us to investigate whether nursing homes in different areas respond heterogeneously to the staffing requirement, splitting

⁴See Cal. Code Regs. Tit. 22, §72038, §72513, §72329, and §72617.

the sample into facilities in ZIP codes that have median incomes, Black population shares, and Hispanic population shares above the median for California ZIP codes.

Appendix Table A.1 reports sample means for the number of beds and staffing level measured using each data source, along with facility characteristics. Our sample is limited to facility-years in which all facility characteristics and the size of the facility are observed. We further exclude facilities in Wisconsin because they face additional staffing requirements if they have at least 100 residents. In total, we have nearly 250,000 facility-year observations from 2000–2019 with over 20,000 of these for facilities in California.

Figure 1 shows the key empirical facts in these data: there is massive bunching in facility size just below the regulatory threshold (Figure 1a), and there is a jump in RN staffing above the regulatory threshold (Figure 1b). The key econometric challenge is that the presence of extensive bunching shown in the first panel means that the data alone cannot determine whether the increase in staffing shown in the second panel is from the treatment effect of the regulation or from the selection effect of low-staffing facilities bunching below the threshold.

3 Methodology

3.1 Setup

In this section, we introduce a simple model of an entity’s decision to avoid or comply with a regulation that applies only above a given threshold in the assignment variable, which in our context is facility size. Entity i chooses assignment variable a_i and the level of the regulated variable r_i , which in our setting is staffing. Utility is determined by how close the entity is to its preferred levels of the assignment variable a_i^* and regulated variable r_i^* :

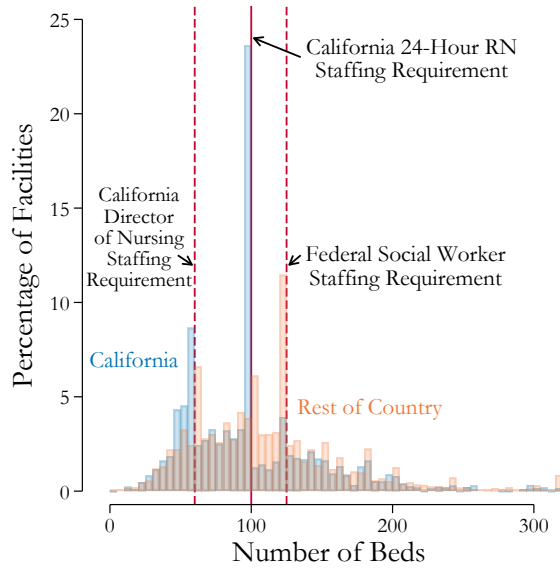
$$U_i = g_i(a_i - a_i^*, r_i - r_i^*)$$

where $g_i()$ is an entity-specific function that is decreasing in $|a_i - a_i^*|$ and $|r_i - r_i^*|$. Hence, entities will choose $a_i = a_i^*$ and $r_i = r_i^*$ in the absence of regulation ($D_i = 0$). On the other hand, in the presence of regulation ($D_i = 1$), entities with an assignment variable greater than a regulatory threshold A must have a value of the regulated variable of at least R , so (a_i^*, r_i^*) may no longer be feasible.

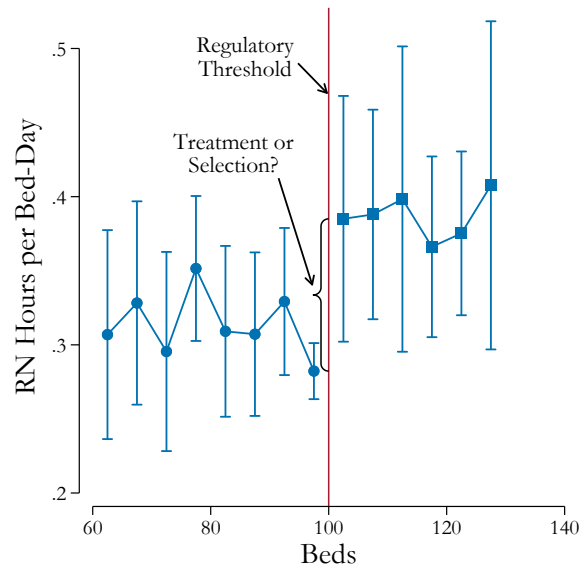
Note that this model is similar to a regression discontinuity (RD) design because the treatment (the size-based regulation) applies only to entities that exceed some threshold. In contrast to standard treatments of RD (McCrary, 2008; Imbens and Lemieux, 2008), we allow entities to manipulate their treatment status, choosing the assignment variable a_i . Given that the assignment variable is manipulable, treatment status D_i will refer to the presence of threshold-based regulation, rather than an entity exceeding the regulatory threshold. Thus, in our context D_i is an indicator for

Figure 1. Number of Facilities and Staffing by Bed Count

(a) Distribution of Nursing Home Sizes in California and in Other States



(b) Staffing Level by California Facility Size



Notes: Panel (a) reports the distribution of nursing home bed counts for California (blue) and all other states (orange) in 5-bed bins. The vertical solid line marks the regulatory threshold, and the vertical dashed lines denote the thresholds above which California facilities are required to have a separate director of nursing services and above which facilities are federally required to employ a full-time social worker. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin. Panel (b) plots binned-scatter means of RN hours per bed-day from the PBJ data against facility size in beds (5-bed bins). The vertical red line denotes the regulatory threshold. The annotated brace denotes the vertical jump at the threshold, which may be caused by treatment or selection. The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics for 2017–2019. The sample is further limited to facilities in California with 60–129 beds. 95% confidence intervals are from standard errors clustered at the facility level.

facility i being in California, the state with the staffing regulation.

To succinctly express entities' hypothetical choices under different regulation regimes, we use potential outcomes notation:

$$a_i = a_i(1) \cdot D_i + a_i(0) \cdot (1 - D_i),$$

$$r_i = r_i(1) \cdot D_i + r_i(0) \cdot (1 - D_i).$$

Similarly, we denote the observed distribution of the assignment variable for the subsample with treatment $D_i = d$ as $f_d(a)$.

In this model, two types of entities choose $a_i(0) = a_i(1) = a_i^*$ and $r_i(0) = r_i(1) = r_i^*$ even in the presence of regulation because their preferred assignment and regulated variable values are admissible under the regulation:

- *Never-regulated*: entities with a preferred assignment variable value small enough that they are not subject to the regulation, or $NeverReg \equiv \{i : a_i^* \leq A\}$;
- *Self-regulated*: entities with a preferred assignment variable value large enough to be subject to the regulation but with a preferred value of the regulated variable that meets the requirement, or $SelfReg \equiv \{i : a_i^* > A, r_i^* \geq R\}$.

In addition, under our model, entities will deviate from a_i^* or r_i^* , but never both, given that the utility function is maximized at $r_i = r_i^*$ regardless of the value of a_i and vice versa.⁵ Furthermore, because utility is decreasing in the distance from the entity's preferred assignment and regulated variable values, entities that deviate in terms of the assignment variable will choose $a_i(1) = A$ and just barely avoid the regulation, and those that deviate in terms of the regulated variable will choose $r_i(1) = R$ and just barely comply with the regulation. This allows us to decompose remaining entities into two types:

- *Bunchers*: entities that reduce their assignment variable value to avoid regulation, or $Buncher \equiv \{i : a_i(1) \leq A < a_i(0) = a_i^*, r_i(0) = r_i(1) = r_i^* < R\}$;
- *Rule-takers*: entities that comply with the regulation by increasing their regulated variable value, or $RuleTaker \equiv \{i : a_i(1) = a_i(0) = a_i^* > A, r_i(1) \geq R > r_i(0) = r_i^*\}$.

In this model, the welfare effects of the regulation come from the changes in behavior by bunchers and rule-takers. Never-regulated and self-regulated entities do not change their behavior

⁵This may seem to be a restrictive assumption, as facilities that downsize may need less staff. However, rather than total staffing, we use staffing hours *per bed-day* as our staffing measure, which is normalized by facility size. We validate that this staffing measure does not meaningfully vary by facility size outside of the manipulation region in California and around 100 beds in other states, as shown by Figure 5 and Appendix Figure A.2.

in response to the regulation, so for social welfare calculations, they are irrelevant. Denoting the social value (ignoring the private costs and benefits) of an entity having assignment variable a_i with regulated variable r_i as $w_i(a_i, r_i)$, the welfare effect of the regulation is

$$\begin{aligned} \mathcal{W} = & \sum_{i \in \{RuleTaker\}} [w_i(a_i^*, R) - w_i(a_i^*, r_i^*)] - [g_i(0, 0) - g_i(0, R - r_i^*)] \\ & + \sum_{i \in \{Buncher\}} [w_i(A, r_i^*) - w_i(a_i^*, r_i^*)] - [g_i(0, 0) - g_i(A - a_i^*, 0)], \quad (1) \end{aligned}$$

where the first bracketed term in each sum is the social value of the entity responding to the regulation and the second is the private cost to the entity.

Equation (1) makes it clear that the welfare effect of the regulation in the presence of avoidance behavior depends on two key objects of interest: the (social and private) effects of avoidance behavior by bunchers, and the effects of compliance by rule-takers. Thus, we will estimate these two objects.

3.2 Bunching Estimation

The model predicts that there will be excess mass just below the threshold and missing mass above the threshold. We refer to the region for which $a_i(0) \neq a_i(1)$ for any i as the manipulation region, and denote it by $\mathcal{M} = (\underline{A}, \bar{A})$. In addition, we partition the manipulation region into the regions below and above the threshold— $\mathcal{M}_{below} \equiv (\underline{A}, A]$ and $\mathcal{M}_{above} \equiv (A, \bar{A})$ —which we call the bunching and missing-mass regions, respectively.

A standard bunching estimator is to fit a polynomial using data for treated entities outside the manipulation region and interpolate. Our setting provides us with a number of opportunities to overcome well-known limitations of these standard methods (Kleven, 2016). In particular, in our setting, we observe the distribution of facility sizes in states not subject to the staffing requirement. Under the assumption that the distributions of preferred facility size within the manipulation region are the same in California as in other states, we can use the size distribution of these untreated facilities as a counterfactual for the size distribution of treated facilities had they not been subject to regulation.⁶ The existence of a control group allows us to relax the functional form assumptions of standard bunching estimators and estimate the counterfactual size distribution non-parametrically, similar to difference-in-discontinuity designs (Kleven et al., 2011; Brown, 2013; Benzarti, 2020). For this reason, we employ standard bunching estimators fitting a polynomial only as a robustness check.

⁶While the assumption of equal distributions within the manipulation region is a strong assumption, one can test whether it is likely to hold by comparing the size distributions of treated and untreated entities outside this region. As we show in Figure 3, this is the case in our setting.

Regardless of how the counterfactual distribution of facility size is obtained, the proportion of buncbers among treated entities in the bunching region is given by the share of mass at the bunching point that is excess mass,

$$\mathbb{P}[Buncher \mid a_i(1) \in \mathcal{M}_{below}] = \frac{(F_1(A) - F_1(\underline{A})) - (F_0(A) - F_0(\underline{A}))}{F_1(A) - F_1(\underline{A})}, \quad (2)$$

and the proportion of buncbers among entities that would counterfactually be in the missing-mass region is given by the share of mass at point a that is missing,

$$\mathbb{P}[Buncher \mid a_i(0) = a, a \in \mathcal{M}_{above}] = \frac{f_0(a) - f_1(a)}{f_0(a)}, \quad (3)$$

as in standard treatments of bunching (Chetty et al., 2011; Kleven and Waseem, 2013; Kleven, 2016).

While bunching estimators usually emphasize excess mass, the implied counterfactual location of buncbers also provides a measure of the intensive size distortion induced by the regulation. We summarize this distortion by estimating the mean preferred size of bunching facilities $\mathbb{E}[a_i(0) \mid Buncher, D_i = 1]$, or in the more compact notation we use throughout the paper, $\bar{a}(0)_{d=1}^{Buncher}$.⁷ While Equation (2) reports *how many* entities bunch, the *degree* to which these entities manipulate their assignment variable is a useful complement for understanding the severity of the avoidance behavior. Although this object follows directly from bunching accounting and is implicitly used in the estimation of structural parameters underlying bunching behavior (Kleven, 2016), to our knowledge, we are the first to report it directly as a transparent, policy-relevant summary of the intensive margin of avoidance.

To obtain the counterfactual average size of bunching facilities, we first estimate the counterfactual size of all facilities in the bunching region. Because untreated facilities provide a valid counterfactual within the manipulation region and facilities do not manipulate their sizes upwards, the observed mean size of control facilities in the manipulation region is equal to a weighted average of the size among (1) the facilities that choose to remain above the threshold in the presence of the regulation and (2) the facilities that choose to be below the threshold. Thus, the counterfactual mean size of facilities in the bunching region satisfies:

$$\bar{a}(0)_{d=1, a \in \mathcal{M}_{below}} = \frac{\bar{a}_{d=0, a \in \mathcal{M}} - (1 - \lambda) \cdot \bar{a}_{d=1, a \in \mathcal{M}_{above}}}{\lambda}, \quad (4)$$

where λ is the share of treated facilities in the manipulation region that are below the threshold.

⁷We write $\bar{Y}_{d,a \in S}^g \equiv \mathbb{E}[Y_i \mid a_i \in S, D_i = d, G_i = g]$ to denote the average value of a variable Y for entities of type $G_i = g$ (e.g., buncbers or never-regulated), with treatment d and assignment variable in set S . We use the same notation for potential outcomes and omit superscripts and subscripts when not conditioning on a specific variable.

With the counterfactual size of all facilities in the bunching region in hand, we can then decompose this object into a weighted average of the mean size of bunchers and never-regulated facilities, with the weights coming from the share of facilities in the bunching region that are bunchers (Equation (2)). Thus, the counterfactual average size of bunchers is

$$\bar{a}(0)_{d=1}^{Buncher} = \frac{\bar{a}(0)_{d=1, a \in \mathcal{M}_{below}} - (1 - \pi) \cdot \bar{a}_{d=0, a \in \mathcal{M}_{below}}}{\pi}, \quad (5)$$

where π is the share of treated facilities in the bunching region that are bunchers.

3.3 Characterizing Bunchers

Next, we characterize bunchers along two dimensions. First, we estimate heterogeneity in the amount of manipulation of the assignment variable among bunchers. To do this, we note that all of the methods discussed in the previous subsection apply conditional on exogenous facility characteristics. Thus, we can use Equations (4) and (5) to identify the degree to which facilities with different baseline characteristics manipulate their size.

Second, we estimate heterogeneity in the propensity of facilities with different characteristics to bunch at all. To do this, we make an assumption on continuity of potential outcomes and facility characteristics, specifically, that outcomes for never-regulated facilities in the treatment group— $\bar{r}(0)_{d=1, \{a\}}^{NeverReg}$ —are smooth throughout for $a \leq A$. This allows us to extrapolate using observations below the manipulation region ($a \leq \underline{A}$) to obtain $\bar{r}(0)_{d=1, a \in \mathcal{M}_{below}}^{NeverReg}$.⁸ This is reminiscent of the assumption requiring continuity of potential outcomes in RD settings (Imbens and Lemieux, 2008), with two differences. First, we allow extrapolation through a region $\mathcal{M}_{below}$ rather than to a single point, and second, in the case where $\mathcal{M}_{below} = \{A\}$ our assumption is slightly weaker in that it only requires left continuity.

This assumption gives us the characteristics of the never-regulated entities at the bunching point. The intuition is that because there is no discrete change in treatment status, any discontinuity in facility characteristics—including potential outcomes—at the bunching point relative to below the bunching region must come from selection (Caetano, 2015; Caetano et al., 2025). With both the share of bunchers and never-regulated entities and the potential outcomes for the never-regulated entities in hand, we can then back out the potential outcomes for the bunchers as well.

Figure 2 shows this identification graphically. The solid points denote observed staffing as a function of facility size in California. The dip at the threshold can be decomposed into average staffing for bunchers (shown by the open triangle) and never-regulated facilities (the open circle), with weights given by the share of bunchers among facilities at the threshold and its complement.

⁸Note that the characteristic here is the potential staffing outcome $r_i(0)$, but the assumption is analogous for any fixed characteristic X_i .

Continuity of potential outcomes allows us to identify average staffing for non-bunchers, which allows us to solve for average staffing among bunchers by rearranging this weighted average:

$$\bar{r}(0)_{d=1}^{Buncher} = \frac{\bar{r}_{d=1,a \in \mathcal{M}_{below}} - (1 - \pi) \cdot \bar{r}(0)_{d=1,a \in \mathcal{M}_{below}}^{NeverReg}}{\pi}. \quad (6)$$

3.4 Estimation of Treatment Effect of Compliance

To identify the treatment effect of compliance with the regulation using our model, we once again decompose counterfactual outcomes for treated facilities with preferred sizes above the threshold (shown by the open square in Figure 2) into components for bunchers (the open triangle) and non-bunchers (the open pentagon), with weights given by the probability of bunching among facilities with preferred size above the threshold.⁹ As shown by the dashed line, we assume that counterfactual outcomes for the combined group can be obtained by smoothly extrapolating outcomes for never-regulated facilities—an extrapolation-type assumption similar in spirit to Arnold et al. (2021, 2022)—while counterfactual outcomes for bunchers are given by Equation (6). Hence, we can rearrange the decomposition to solve for counterfactual outcomes among non-bunchers (rule-takers and self-regulated facilities) in the missing-mass region:

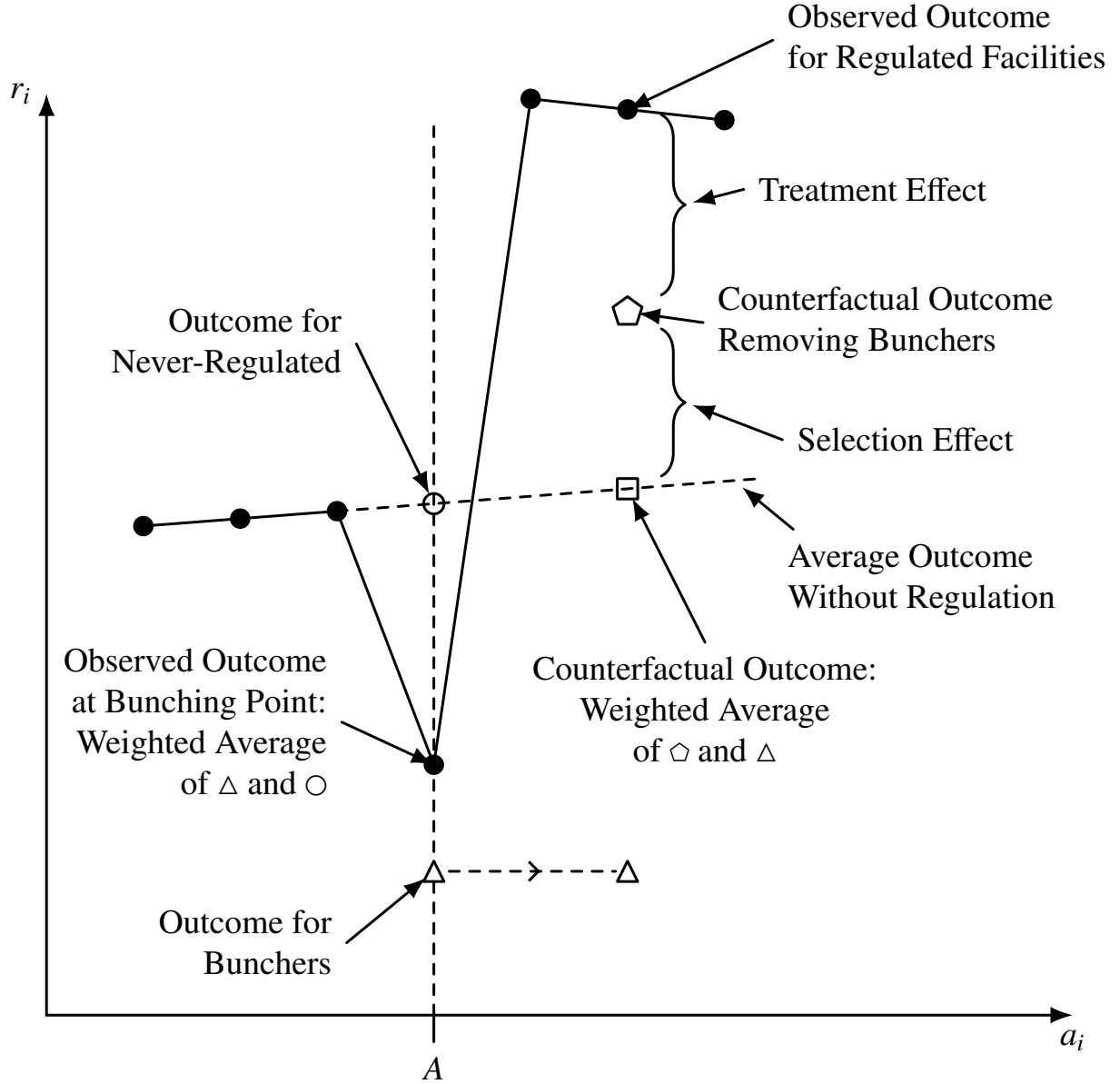
$$\bar{r}(0)_{d=1,a \in \mathcal{M}_{above}} = \frac{\bar{r}(0)_{d=1,a(0) \in \mathcal{M}_{above}} - \psi \cdot \bar{r}(0)_{d=1}^{Buncher}}{1 - \psi}, \quad (7)$$

where ψ is the share of facilities with preferred bed sizes in the missing-mass region that are bunchers. The first term in the numerator is the counterfactual mean of the regulated variable among all facilities that would be in the missing-mass region in the absence of regulation (bunchers, rule-takers, and self-regulated facilities), while the second term is the product of the share of this group that is bunchers (from Equation (3)) and the counterfactual mean value of the regulated variable among bunchers (from Equation (6)). Thus, this equation formalizes the idea that we are identifying the counterfactual outcomes of those that choose to remain above the regulatory threshold by “subtracting off” the counterfactual outcome for bunchers from the counterfactual outcome among all facilities.

With the counterfactual outcomes for those that comply with the regulation in hand, the average effect of the regulation on the facilities subject to the regulation (those that remain in the missing-mass region, i.e., rule-takers and self-regulated facilities), $TOT \equiv \bar{r}(1)_{d=1,a \in \mathcal{M}_{above}} - \bar{r}(0)_{d=1,a \in \mathcal{M}_{above}}$,

⁹Note that it is possible to derive bounds for the effect in a fully reduced-form way without using our model, as described in Gerard et al. (2020). Unless manipulation is modest, these bounds tend to be wide, as is the case in our setting. Comparing staffing above the regulatory threshold with the trimmed distributions of staffing just below the threshold, we obtain bounds for the treatment effect on RN hours per bed-day of $[-0.243, 0.297]$, which is extremely wide given that mean staffing levels in California are 0.317.

Figure 2. Identification of Potential Outcomes



Notes: Figure is a stylized illustration of the identification strategy described in Sections 3.3 and 3.4. Solid points denote hypothetical observed staffing $\mathbb{E}[r_i | a_i]$ as a function of facility size for California ($D_i = 1$); the vertical dashed line marks the regulatory threshold A ; and the sloped dashed line labeled “Average Outcome Without Regulation” extrapolates the linear trend among never-regulated facilities through and beyond the threshold. Open shapes represent hypothetical counterfactual means: the open circle is the mean for never-regulated facilities at the bunching point, which under a flat trend coincides in height with the open square; the open triangle is the mean for bunchers; the open square is the pooled counterfactual mean for all facilities that would be in the missing-mass region absent the regulation (bunchers, rule-takers, and self-regulated facilities); and the open pentagon is the counterfactual mean for facilities that remain above the threshold, removing bunchers. The open triangle representing implied staffing for bunchers is extended horizontally from the bunching region to the missing-mass region to illustrate that the open square is a weighted average of the open triangle and pentagon. The upper brace marks the treatment effect of the regulation (observed outcome for regulated facilities less their counterfactual outcome); the lower brace marks the selection effect induced by bunching (the counterfactual mean rises when negatively selected bunchers are removed from the pool).

is given by the difference between observed and counterfactual outcomes for these facilities:

$$TOT = \bar{r}_{d=1, a \in \mathcal{M}_{above}} - \bar{r}(0)_{d=1, a \in \mathcal{M}_{above}}. \quad (8)$$

Derivations of all formulae are given in Appendix B, and for all estimates in the next section, we report standard errors from 2,000 bootstrap iterations clustered at the facility level unless otherwise noted.

4 Results

4.1 Extent of Bunching

Figure 3 shows the distribution of nursing home sizes in California $f_1(a)$ (in blue), and in other states $f_0(a)$ (in orange). With the exception of excess mass in California just below the threshold above which California facilities are required to have a separate director of nursing, the size distributions in California and other states leading up to the threshold are very similar. This suggests that untreated states may serve as a reasonable counterfactual distribution for California. On the other hand, we observe a striking amount of excess mass at the threshold in California, with no similar spike in other states. In particular, 19% of California facilities have 99 beds, over 17 times the share in other states. This difference is highly statistically significant ($p < 0.001$), with California also having statistically significantly more facilities at 98 beds and 97 beds ($p < 0.05$), although these differences are much more modest.¹⁰ For this reason, we treat 97–99 beds as the bunching region. Using Equation (2), we estimate that the share of nursing homes in California in the bunching region that are bunchers is 89.3%.¹¹ Figure 4a reports the share of bunchers at each size from 95 to 99 beds.

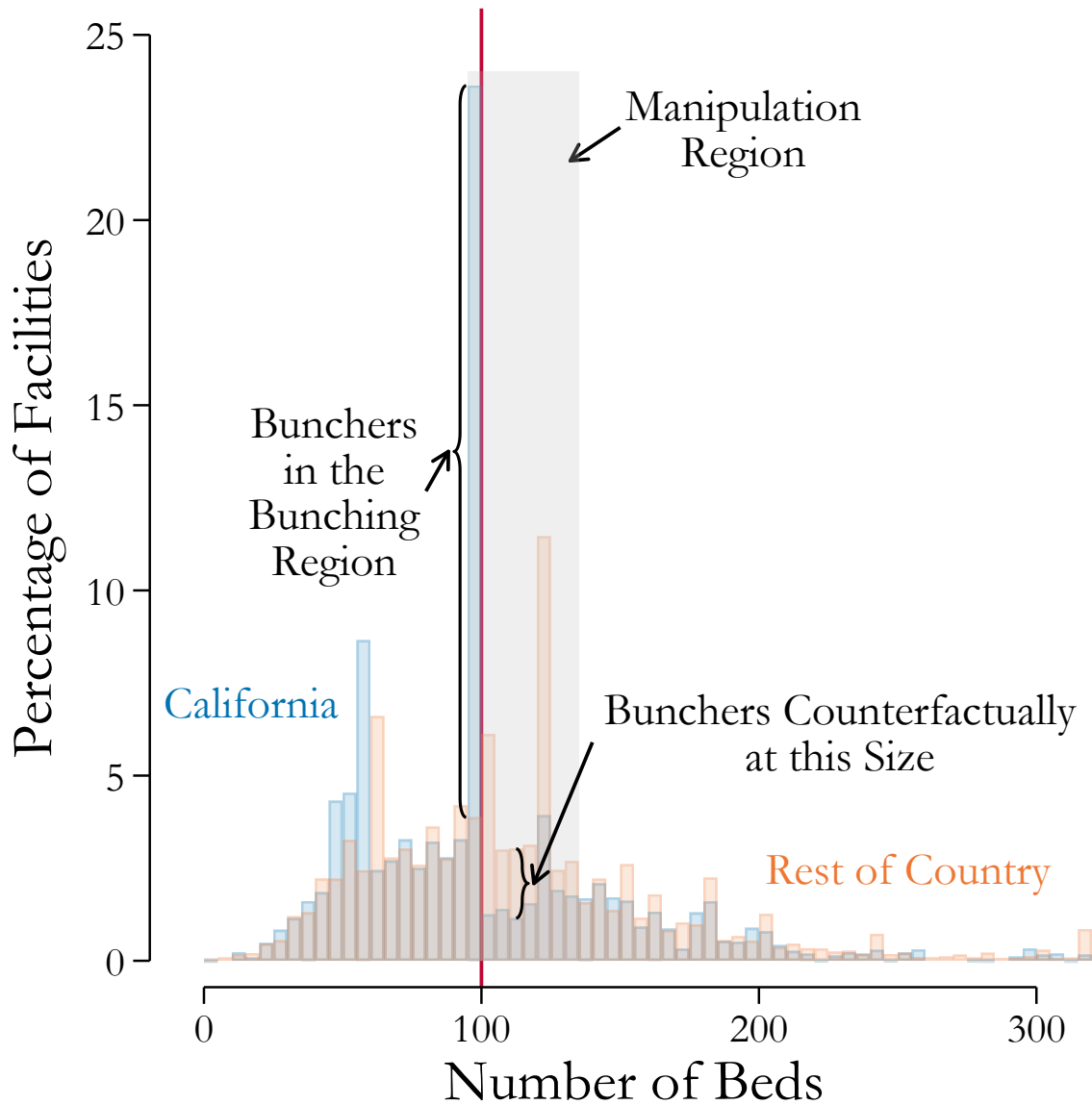
Above the threshold, we observe missing mass in California, with $f_1(a) < f_0(a)$ until 135 beds, at which point $f_1(a) \approx f_0(a)$ once again, suggesting this is the point at which the manipulation region ends. The amount of missing mass decreases meaningfully after the threshold for the federal requirement to employ a social worker, consistent with few facilities that would otherwise comply with the social worker requirement manipulating their size to avoid the 24-hour RN requirement.

More formally, we use Equation (3) to compute the share of bunchers as a function of preferred (counterfactual) bed sizes in California in the absence of regulation, with the share of mass that is missing giving the share of facilities at that preferred size that are bunchers, as marked on Figure 3. We estimate that nearly 80% of facilities with 100–104 beds would prefer to bunch rather than comply with the staffing requirement, with this percentage remaining above 50% through the 120–

¹⁰California facilities are 145% (137%) more likely to have 97 (98) beds.

¹¹The standard error of this estimate is 0.80%.

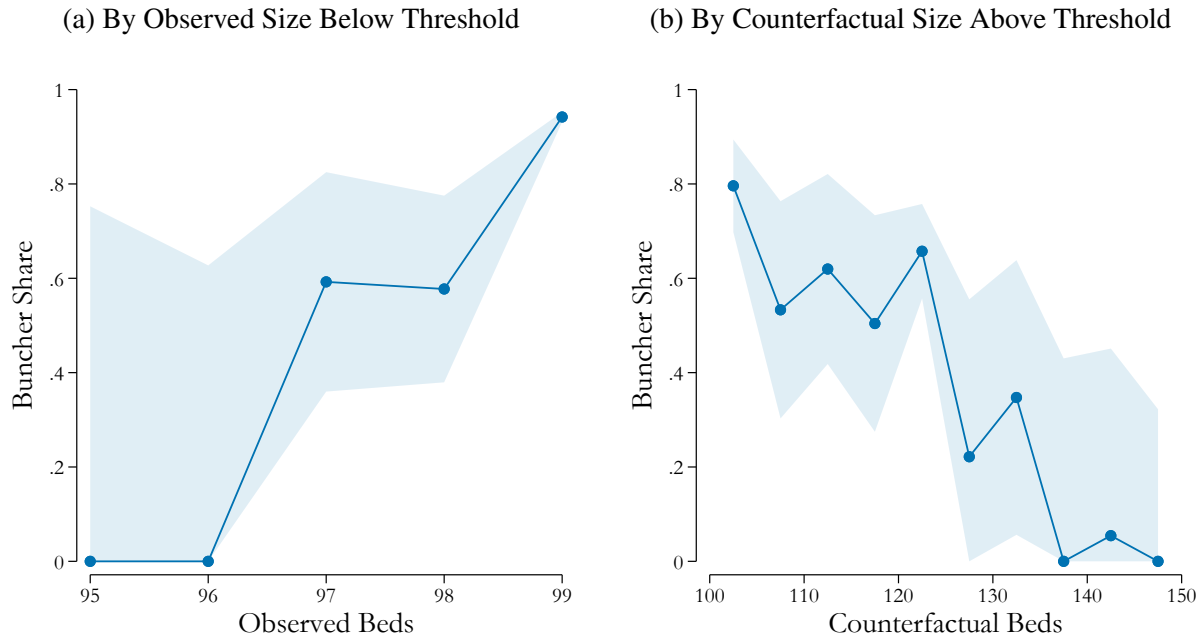
Figure 3. Distribution of Nursing Home Sizes in California and in Other States Showing Excess and Missing Mass



Notes: Figure reports the distribution of nursing home bed counts for California (blue) and all other states (orange) in 5-bed bins. The gray shaded region denotes the manipulation region. The excess mass in the bunching region and the density shortfall in the missing-mass region are highlighted by braces. The vertical solid line marks the regulatory threshold. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin.

124 bed bin. Beyond 135 beds, the estimated share of bunchers is not statistically different from zero. These estimates are reported in Figure 4b. Appendix C shows that results are similar if we instead estimated the counterfactual distribution using traditional methods.

Figure 4. Share of Facilities that are Bunchers



Notes: Panel (a) reports the estimated share of California nursing homes at each observed bed count that are bunchers, computed as the difference between the density in California and other states, divided by the density in California following Equation (2). Shares are reported for 95–99 beds. Panel (b) reports the estimated share of California nursing homes at each counterfactual bed count that are bunchers, computed following Equation (3), in 5-bed bins. Shares are reported for 100–149 beds. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin. 95% confidence intervals are from standard errors clustered at the facility level.

Not only do many facilities choose to bunch to avoid the staffing regulation, but bunching results in large changes in facility size. Using Equation (4) and a manipulation region of $\mathcal{M} = (96, 135)$, we estimate that facilities in the bunching region would counterfactually have had 112.1 beds. For bunchers specifically, the average counterfactual size is 113.8 beds,¹² meaning bunching facilities shrink by over 13%, on average. The average counterfactual size of bunchers is smaller than the average size of facilities in the missing-mass region that do not bunch (rule-takers and self-regulated facilities), 118.7 beds, which indicates that bunchers are negatively selected on size, consistent with bunchers being facilities for which staying below the size threshold is less costly.

¹²This estimate has a standard error of 0.59 beds.

4.2 Characteristics of Bunchers

Having demonstrated that there is extensive bunching in this context, we next turn to characterizing bunchers along observable characteristics. Table 1 reports heterogeneity in the propensity of facilities to bunch as well as in the extent to which facilities manipulate their size to avoid the staffing requirement.

Panel A of Table 1 characterizes bunchers and non-bunchers (i.e., never-regulated facilities) just below the threshold, using methods from Section 3.3. To extrapolate baseline characteristics of never-regulated facilities to the manipulation region below the threshold, we estimate a local linear fit with MSE-optimal bandwidth (Imbens and Kalyanaraman, 2012; Calonico et al., 2014; Gelman and Imbens, 2019). We find that bunchers are somewhat more likely to be for-profit and chain-owned than non-bunchers, but these differences are not statistically significant, with no meaningful difference in the probability the facility has an Alzheimer’s special care unit. In terms of the ZIP code of the facility, bunchers are nearly twice as likely as non-bunching facilities to be in below-median income areas and more than twice as likely to be in above-median Hispanic population share areas. In Appendix Table A.2, we show that we get broadly similar results estimating Equation (2) using the conditional size distribution for facilities with each characteristic in other states as the counterfactual.

Panel B of Table 1 shows that there is limited heterogeneity in the *degree* of size manipulation by bunching facilities by facility characteristics, with none of the differences being statistically significant. Given the similar degree of bunching and the greater propensity to bunch, these results indicate that not only does the staffing requirement reduce facility size, but also that this reduction in access is most acute in disadvantaged areas.

4.3 Effect on Staffing

Next, we estimate the effect of the regulation on staffing following Subsection 3.4. Figure 5 shows RN hours per bed-day in the PBJ data as a function of facility size.¹³ Notice the similarity with Figure 2, where staffing among facilities at the bunching point is significantly below the average among those below the bunching region. At the same time, above the threshold, staffing is higher in the manipulation region before falling back to the pre-manipulation region mean. These patterns suggest, as predicted by the model, that facilities with low staffing (and thus the highest necessary change in staffing) are disproportionately bunching below the 100-bed threshold. We observe no such patterns in states other than California.¹⁴

In addition, we observe in Figure 5 that staffing is roughly constant as a function of facility size

¹³Appendix Figure A.1 and Table A.3 show we get very similar results using staffing data from OSCAR or HCRIS.

¹⁴See Appendix Figure A.2.

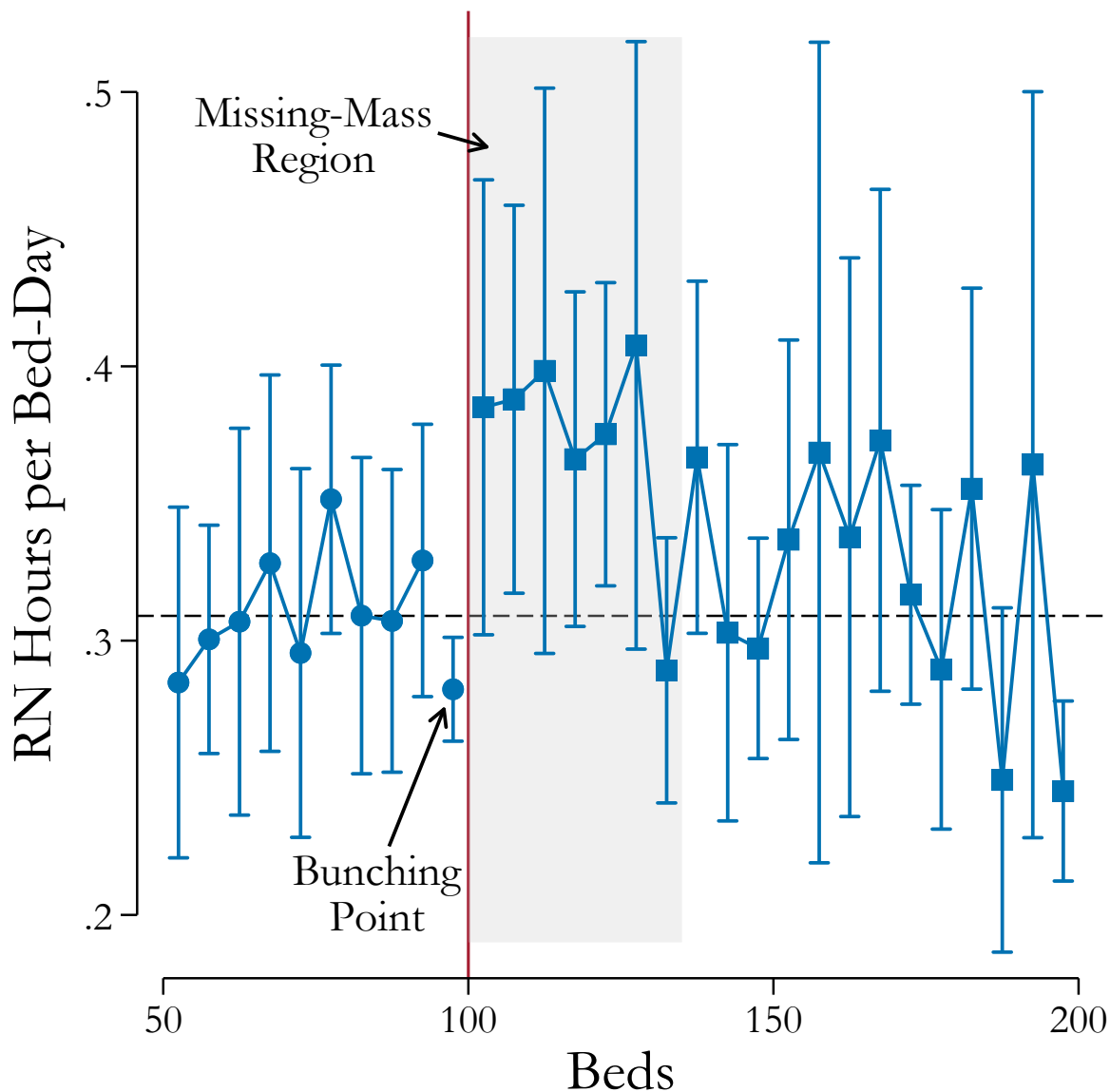
Table 1. Characterizing Bunchers and Non-Bunchers

<i>Panel A: Share of Facilities</i>				
Characteristic	Bunchers	Non-Bunchers	Difference	SE of Diff.
<i>Facility Characteristics</i>				
For-Profit	0.919	0.825	0.094	0.065
Chain Ownership	0.531	0.493	0.038	0.092
Has Alz. Unit	0.056	0.063	-0.007	0.030
<i>ZIP Characteristics</i>				
Low Income	0.667	0.349	0.318***	0.121
High Black	0.532	0.481	0.051	0.104
High Hispanic	0.639	0.279	0.360***	0.113
<i>Panel B: Counterfactual Size of Bunchers</i>				
Characteristic	Has Characteristic	Does Not Have Characteristic	Difference	SE of Diff.
<i>Facility Characteristics</i>				
For-Profit	113.8	114.3	-0.53	1.27
Chain Ownership	114.2	113.3	0.91	0.87
Has Alz. Unit	112.4	113.8	-1.33	2.51
<i>ZIP Characteristics</i>				
Low Income	113.5	114.8	-1.34	1.24
High Black	113.4	114.1	-0.65	1.14
High Hispanic	115.6	113.7	1.96	1.28

Notes: Panel A reports the share of bunchers and non-bunchers (never-regulated facilities) in the bunching region (97–99 beds) with each characteristic, estimated by applying Equation (6) with the characteristic X_i in place of staffing r_i . Panel B reports counterfactual size in beds for bunchers in each characteristic group, estimated using Equation (5). ZIP characteristics “Low Income,” “High Black,” and “High Hispanic” indicate that the facility’s ZIP code is below or above the median for California ZIP codes. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin. The sample in panel B is further limited to facilities with 97–134 beds. Standard errors are from 2,000 bootstrap iterations at the facility level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure 5. Staffing Level by California Facility Size with Bunching Point and Missing-Mass Region

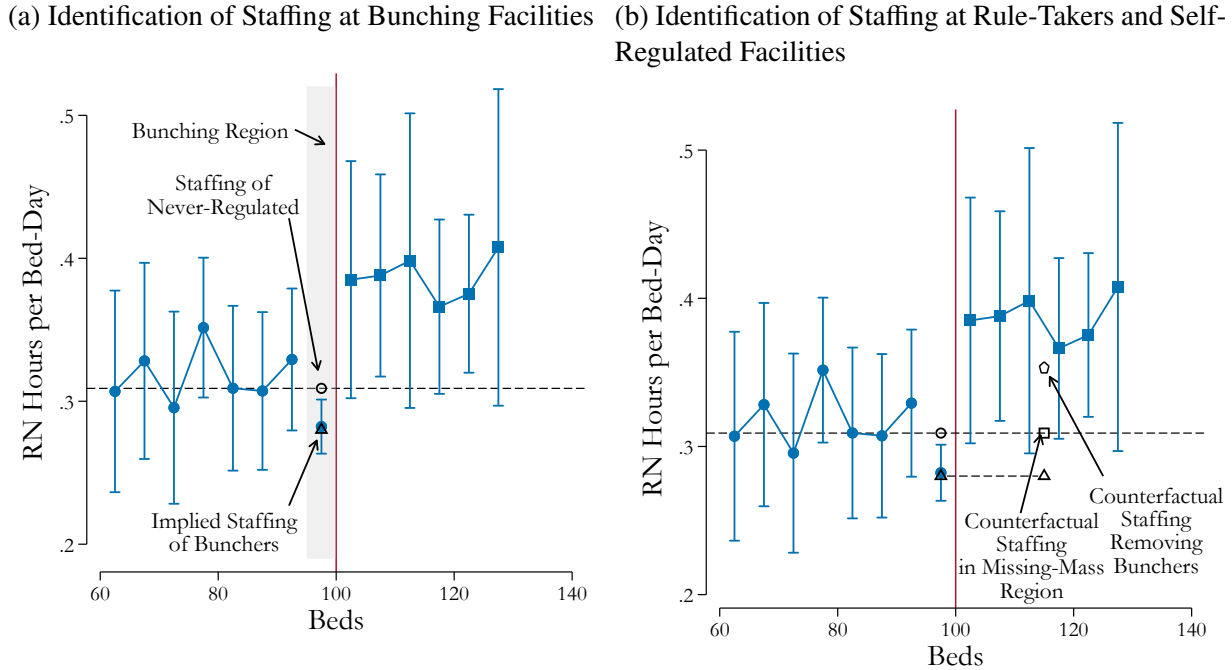


Notes: Figure plots binned-scatter means of RN hours per bed-day from the PBJ data against facility size in beds (5-bed bins). The vertical red line denotes the regulatory threshold, and the bunching point is shown immediately to the left. The gray shaded region denotes the missing-mass region. The horizontal dashed line denotes the mean staffing level for facilities with 80–96 beds. The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics for 2017–2019. The sample is further limited to facilities in California with 50–199 beds. 95% confidence intervals are from standard errors clustered at the facility level.

outside the manipulation region. Therefore, to simplify estimation we strengthen the smoothness assumption of Section 3.3 and assume that staffing is mean-independent of preferred size conditional on treatment status ($\bar{r}(0)_{d=1,a(0) \in \mathcal{M}_{above}} = \bar{r}_{d=1,a \leq \underline{A}}$). Thus, we assume the counterfactual mean staffing at all (counterfactual) facility sizes is the same as that of the never-regulated, which we estimate using facilities with 80–96 beds: 0.309 RN hours per bed-day.

Using Equation (6), we find that mean staffing among bunchers $\bar{r}(0)_{d=1}^{Buncher}$ is 0.280, 9.4% lower than among never-regulated facilities, indicating that bunchers are negatively selected on staffing level. Figure 6 mirrors Figure 2 by adding our estimates of the unobserved quantities denoted by shapes to the plot of observed staffing levels. Figure 6a shows the implied average staffing level of bunchers (the open triangle) compared to that of never-regulated facilities (the open circle).

Figure 6. Staffing Level by California Facility Size Showing Identification of Unobserved Staffing Levels



Notes: Panels (a) and (b) plot binned-scatter means of RN hours per bed-day from the PBJ data against facility size in beds (5-bed bins). The vertical red line denotes the regulatory threshold. The horizontal dashed line denotes the mean staffing level for facilities with 80–96 beds. Panel (a) shows the bunching region shaded in gray, along with the implied staffing of never-regulated and bunching facilities displayed as an open circle and open triangle, respectively. Panel (b) additionally plots the implied counterfactual staffing absent the regulation in the missing-mass region with and without bunchers as an open square and pentagon, respectively. The open triangle representing implied staffing for bunchers is extended horizontally from the bunching region to the missing-mass region to illustrate that the open square is a weighted average of the open triangle and pentagon. The sample in both panels is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics for 2017–2019. The sample is further limited to facilities in California with 60–129 beds. 95% confidence intervals are from standard errors clustered at the facility level.

Having in hand the overall counterfactual mean staffing level (the open square in Figures 2 and 6b), the share of facilities that bunch, and the staffing level of bunching facilities (the open triangle), we can then calculate the counterfactual staffing level of facilities subject to the regulation that do not bunch (rule-takers and self-regulated facilities) using Equation (7). We find that the counterfactual mean staffing among compliant facilities in the missing-mass region $\bar{r}(0)_{d=1, a \in \mathcal{M}_{above}}$ is 0.353 RN hours per bed-day. This is denoted by the open pentagon in Figure 6b and is only slightly lower than the mean staffing of 0.371 among facilities subject to the regulation. Therefore, the estimated treatment effect on the treated TOT is only 0.018 and not statistically different from zero.

Our estimated treatment effect is much smaller than a naïve comparison of staffing above and below the threshold would indicate. The raw difference between facilities subject to the regulation and those with 80–96 beds is 0.062, and estimating a local linear RD that ignores precise manipulation by facilities yields an estimate of 0.182,¹⁵ an estimate 10 times as large as ours.

Table 2 summarizes these results. We find that bunching facilities have lower counterfactual staffing levels than non-bunching facilities, such that nearly the entire observed difference in staffing levels among facilities subject to the staffing requirement is due to selection rather than treatment. Thus, while the staffing regulation causes facilities to manipulate their size to avoid it, it has only a negligible effect on staffing.

4.4 Welfare Effects

Finally, with the key objects of interest—the effects of avoidance behavior by bunchers and compliance by rule-takers—in hand, we return to Equation (1) to assess the overall welfare effect of the staffing policy. Because the estimated treatment effect on staffing is null, so are the welfare effects of rule-takers’ compliance behavior, either because there are very few rule-takers, because compliance requires very little change in staffing for these facilities, or both. Thus, the welfare effects of the policy depend entirely on the welfare effects of avoidance behavior by bunchers.

The welfare effect of bunching in our context is likely to be negative and, we argue, significantly so. This is for two main reasons. First, the model presented in Section 3.1 guarantees that bunching imposes private costs on bunching facilities: avoidance behavior requires deviating from facilities’ preferred number of beds. Second, the avoidance behavior likely has negative social consequences given the degree to which it restricts access to nursing homes in disadvantaged areas. Given the well-established preference of nursing home patients for facilities very close to their homes (Hackmann, 2019; Gandhi, 2023; Einav et al., 2025), the reduction in access in low-income, high-Hispanic ZIP codes we estimate is likely very costly to patients, with its incidence

¹⁵We use an MSE-optimal bandwidth and inference and bias corrections from Calonico et al. (2014). The RD estimate is larger than the raw difference since the latter excludes the bunching region.

Table 2. Effect on Staffing

Parameter	Expression	Estimate
Counterfactual Staffing of All Facilities	$\bar{r}(0)_{d=1}$	0.309 (0.0145)
Observed Staffing in Missing-Mass Region	$\bar{r}(1)_{d=1, a \in \mathcal{M}_{above}}$	0.371 (0.0152)
Staffing of Bunchers	$\bar{r}(0)_{d=1}^{Buncher}$	0.280 (0.0114)
Counterfactual Staffing in Missing-Mass Region	$\bar{r}(0)_{d=1, a \in \mathcal{M}_{above}}$	0.353 (0.0429)
Treatment Effect on the Treated	$\bar{r}(1)_{d=1, a \in \mathcal{M}_{above}} - \bar{r}(0)_{d=1, a \in \mathcal{M}_{above}}$	0.018 (0.0460)

Notes: Estimates of treatment-effect parameters from Section 3.4, using manipulation region (96, 135) and bunching region [97, 99]. Staffing is RN hours per bed-day from the PBJ data. The counterfactual mean staffing of all facilities is identified by the mean-independence assumption stated in Section 4.3; bunchers' counterfactual staffing is from Equation (6); the counterfactual staffing of treated facilities is from Equation (7); the treatment effect on the treated (*TOT*) is from Equation (8). The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics for 2017–2019, excluding facilities in Wisconsin. The sample is further limited to facilities with 80–134 beds. Standard errors from 2,000 bootstrap iterations at the facility level are reported in parentheses.

on disadvantaged communities only heightening this cost.

5 Conclusion

This paper highlights how threshold-based regulations can be rendered ineffective when regulated entities can manipulate their treatment status. Using a simple, novel empirical framework, we document how California's 100-bed staffing rule led to widespread bunching of nursing homes at 99 beds, with the reduction in access being most severe in disadvantaged areas. Furthermore, because facilities that would otherwise have the lowest staffing manipulate their size to avoid the regulation, the requirement had virtually no effect on average staffing, illustrating how avoidance behavior can undermine the effectiveness of threshold-based regulations.

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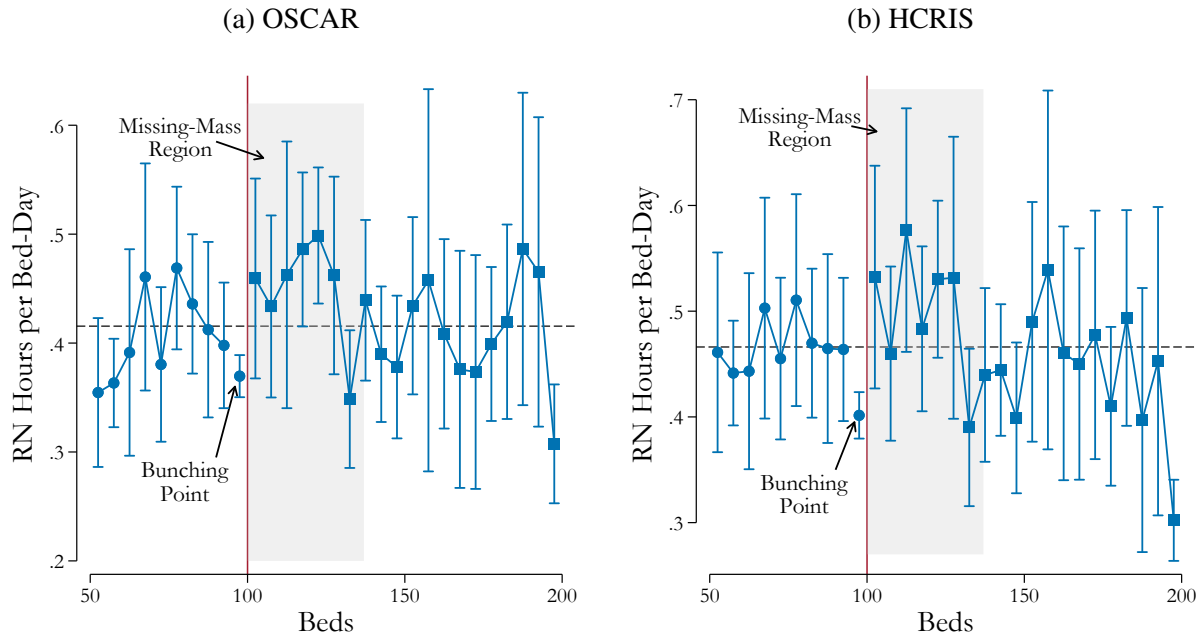
A Appendix Figures and Tables

Table A.1. Summary Statistics

	California	Other States
<i>Facility Size</i>		
Beds	100.5	111.4
<i>Staffing Measures</i>		
RN Hours per Bed-Day (PBJ)	0.317	0.329
RN Hours per Bed-Day (OSCAR)	0.397	0.354
RN Hours per Bed-Day (HCRIS)	0.453	0.442
<i>Facility Characteristics</i>		
For-Profit	0.840	0.692
Chain Ownership	0.522	0.564
Has Alzheimer's Unit	0.063	0.182
<i>ZIP Characteristics</i>		
Median Income (thousands)	29.12	26.26
Black Share	0.061	0.120
Hispanic Share	0.359	0.098
Observations	20,494	227,980

Notes: Sample means of facility size, staffing measures, facility characteristics, and ZIP characteristics for California vs. other states. An observation is a facility-year. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin. PBJ staffing data are available only for 2017–2019, OSCAR staffing data from 2011–2017, and HCRIS staffing data from 2011–2019.

Figure A.1. Staffing Level in Other Sources by California Facility Size



Notes: Panel (a) plots binned-scatter means of RN hours per bed-day from the OSCAR data against facility size in beds (5-bed bins). Panel (b) Panel (b) plots the same using the HCRIS data. The vertical red line denotes the regulatory threshold, and the bunching point is shown immediately to the left. The gray shaded region denotes the missing-mass region. The horizontal dashed line denotes the mean staffing level for facilities with 80–96 beds. The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics. The sample is further limited to facilities in California with 50–199 beds and in panel (a) to 2011–2017 and in panel (b) to 2011–2019. 95% confidence intervals are from standard errors clustered at the facility level.

Table A.2. Share of Bunchers in the Bunching Region by Characteristics

Facility Characteristic	Has Characteristic	Does Not Have Characteristic	Difference	SE of Difference
For-Profit	0.896	0.864	0.032**	0.016
Chain Ownership	0.885	0.897	-0.013	0.011
Has Alz. Unit	0.886	0.889	-0.003	0.022
Low Income	0.881	0.913	-0.032**	0.013
High Black	0.904	0.869	0.034**	0.014
High Hispanic	0.937	0.881	0.057***	0.017

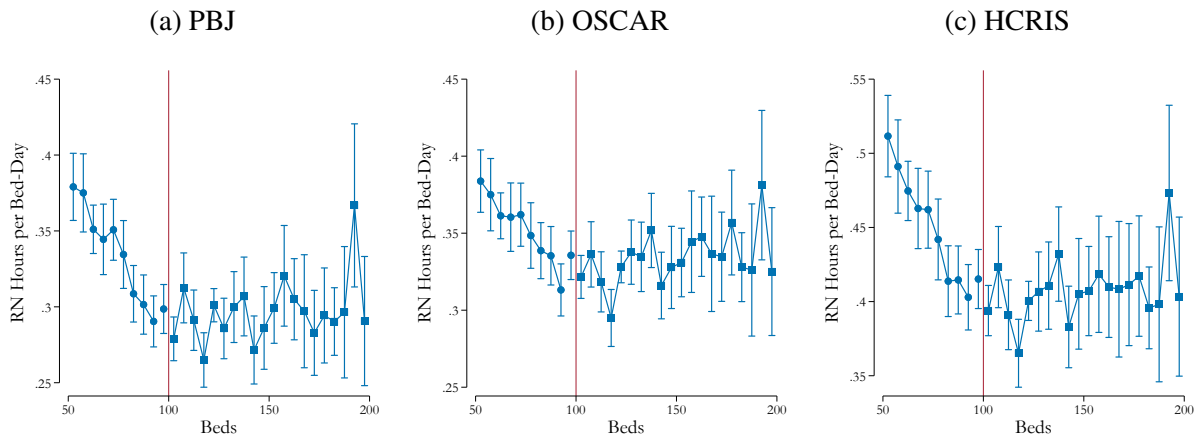
Notes: Table reports the share of bunchers in the bunching region (97–99 beds) for each characteristic group, estimated using Equation (2) with the sample limited to facilities with or without the relevant characteristic. ZIP characteristics “Low Income,” “High Black,” and “High Hispanic” indicate that the facility’s ZIP code is below or above the median for California ZIP codes. The sample is limited to facility-years with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019, excluding facilities in Wisconsin. The sample is further limited to facilities with 97–134 beds. Standard errors are from 2,000 bootstrap iterations at the facility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.3. Effect on Staffing, Alternative Staffing Data

Parameter	Estimate	Standard Error
<i>Panel A: Staffing from OSCAR</i>		
Counterfactual Staffing of All Facilities	0.408	0.0186
Observed Staffing in Missing-Mass Region	0.458	0.0165
Staffing of Bunchers	0.366	0.0116
Counterfactual Staffing in Missing-Mass Region	0.471	0.0540
Treatment Effect on the Treated	-0.014	0.0571
<i>Panel B: Staffing from HCRIS</i>		
Counterfactual Staffing of All Facilities	0.459	0.0207
Observed Staffing in Missing-Mass Region	0.503	0.0193
Staffing of Bunchers	0.394	0.0128
Counterfactual Staffing in Missing-Mass Region	0.560	0.0605
Treatment Effect on the Treated	-0.057	0.0641

Notes: Estimates of treatment-effect parameters from Section 3.4, using manipulation region (96, 135) and bunching region [97, 99]. Panel A reports RN hours per bed-day from the OSCAR data. Panel B reports RN hours per bed-day from the HCRIS data. The counterfactual mean staffing of all facilities is identified by the mean-independence assumption stated in Section 4.3; bunchers’ counterfactual staffing is from Equation (6); the counterfactual staffing of treated facilities is from Equation (7); the treatment effect on the treated (TOT) is from Equation (8). The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics, excluding facilities in Wisconsin. The sample is further limited to facilities with 80–134 beds and in panel A to 2011–2017 and in panel B to 2011–2019. Standard errors from 2,000 bootstrap iterations at the facility level are also reported.

Figure A.2. Staffing Level by Facility Size in Other States



Notes: Panel (a) plots binned-scatter means of RN hours per bed-day from the PBJ data against facility size in beds (5-bed bins) for facilities in states other than California. Panels (b) and (c) plot the same using the OSCAR and HCRIS data, respectively. The vertical red line denotes the regulatory threshold in California. The sample is limited to facility-years with non-missing staffing, size, facility characteristics, and ZIP characteristics. The sample is further limited to facilities in states other than California and Wisconsin with 50–199 beds and in panel (a) to 2017–2019, in panel (b) to 2011–2017, and in panel (c) to 2011–2019. 95% confidence intervals are from standard errors clustered at the facility level.

B Derivations

The following steps show the derivation of Equation (4).

$$\begin{aligned}
\bar{a}_{d=0,a \in \mathcal{M}} &= \bar{a}(0)_{d=0,a \in \mathcal{M}} \\
&= \bar{a}(0)_{d=1,a(0) \in \mathcal{M}} \\
&= \bar{a}(0)_{d=1,a \in \mathcal{M}} \\
&= \lambda \cdot \bar{a}(0)_{d=1,a \in \mathcal{M}_{below}} + (1 - \lambda) \cdot \bar{a}(0)_{d=1,a \in \mathcal{M}_{above}} \\
&= \lambda \cdot \bar{a}(0)_{d=1,a \in \mathcal{M}_{below}} + (1 - \lambda) \cdot \bar{a}_{d=1,a \in \mathcal{M}_{above}}
\end{aligned}$$

Rearranging then yields Equation (4). The first equality comes from the definition of potential outcomes. The second equality comes from the assumption that the distribution of a_i in the manipulation region among untreated units serves as a valid counterfactual for the distribution of a_i absent the regulation among regulated units. The third equality comes from the definition of the manipulation region. The fourth equality follows by the law of total expectation, and the fifth equality follows from the model's prediction that units do not manipulate a_i upwards in response to the regulation.

The following steps show the derivation of Equation (5).

$$\begin{aligned}
\bar{a}(0)_{d=1,a \in \mathcal{M}_{below}} &= \pi \cdot \bar{a}(0)_{d=1,a \in \mathcal{M}_{below}}^{Buncher} + (1 - \pi) \cdot \bar{a}(0)_{d=1,a \in \mathcal{M}_{below}}^{\neg Buncher} \\
&= \pi \cdot \bar{a}(0)_{d=1}^{Buncher} + (1 - \pi) \cdot \bar{a}(0)_{d=1,a(0) \in \mathcal{M}_{below}} \\
&= \pi \cdot \bar{a}(0)_{d=1}^{Buncher} + (1 - \pi) \cdot \bar{a}_{d=0,a \in \mathcal{M}_{below}}
\end{aligned}$$

Rearranging then yields Equation (5). The first equality comes from the law of total expectation. The second comes from the definition of bunchers. The third comes from the assumption that the distribution of a_i in the manipulation region among untreated units serves as a valid counterfactual for the distribution of a_i absent the regulation among regulated units and the definition of potential outcomes. Note that $\bar{a}(0)_{d=1,a \in \mathcal{M}_{below}}$ is identified by Equation (4) and π is identified by Equation (2).

The following steps show the derivation of Equation (6).

$$\begin{aligned}
\bar{r}_{d=1,a \in \mathcal{M}_{below}} &= \bar{r}(0)_{d=1,a \in \mathcal{M}_{below}} \\
&= \pi \cdot \bar{r}(0)_{d=1,a \in \mathcal{M}_{below}}^{Buncher} + (1 - \pi) \cdot \bar{r}(0)_{d=1,a \in \mathcal{M}_{below}}^{\neg Buncher} \\
&= \pi \cdot \bar{r}(0)_{d=1}^{Buncher} + (1 - \pi) \cdot \bar{r}(0)_{d=1,a \in \mathcal{M}_{below}}^{NeverReg}.
\end{aligned}$$

Rearranging yields Equation (6). The first equality follows from the assumption that bunchers do not manipulate r_i and the definition of potential outcomes for the never-regulated facilities. The second equality comes from the law of total expectation. The third equality follows by the definition of bunchers and never-regulated facilities. Note that the derivation holds for any characteristic X_i (i.e., not just r_i) that is not affected by bunching.

The following steps show the derivation of Equation (7).

$$\begin{aligned}\bar{r}(0)_{d=1, a(0) \in \mathcal{M}_{above}} &= \psi \cdot \bar{r}(0)_{d=1, a(0) \in \mathcal{M}_{above}}^{Buncher} + (1 - \psi) \cdot \bar{r}(0)_{d=1, a(0) \in \mathcal{M}_{above}}^{-Buncher} \\ &= \psi \cdot \bar{r}(0)_{d=1}^{Buncher} + (1 - \psi) \cdot \bar{r}(0)_{d=1, a \in \mathcal{M}_{above}}.\end{aligned}$$

Rearranging yields Equation (7). The first equality follows from the law of total expectation. The second equality follows from the definition of bunchers, rule-takers, and self-regulated facilities. Note that the buncher share ψ is identified by Equation (3); the bunchers' counterfactual staffing $\bar{r}(0)_{d=1}^{Buncher}$ is identified by Equation (6). Thus, having in hand the mean counterfactual outcome (absent the regulation) among facilities with preferred size in the missing-mass region—which in our setting is equal to the average staffing below the manipulation region, but in other contexts may be an extrapolation or the average in non-treated jurisdictions or time periods—delivers the mean counterfactual outcome among the treated.

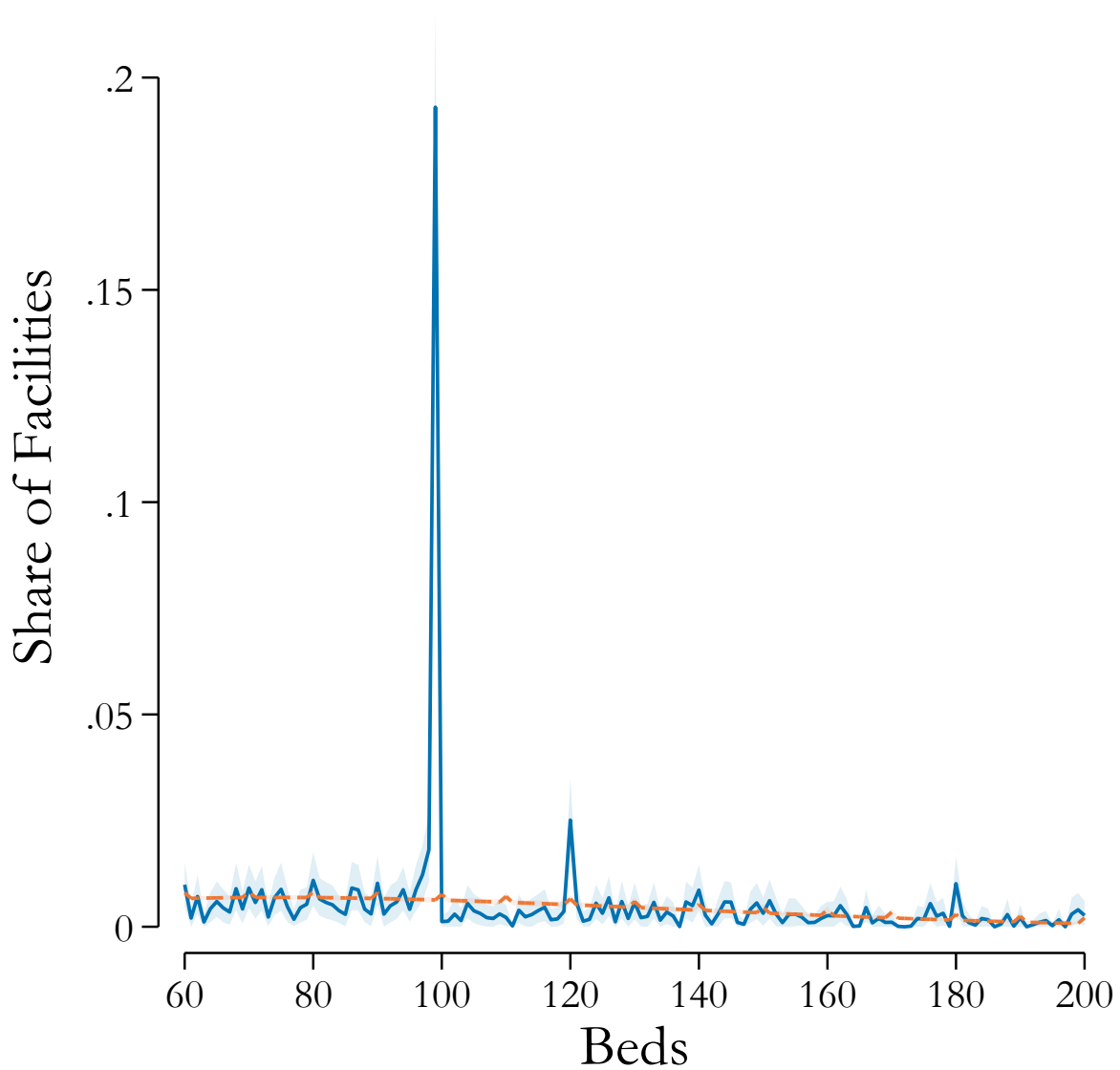
C Results Using Traditional Interpolation Methods

We use data outside a conservative manipulation window of $[90, 150)$ in California to estimate a K th-order polynomial allowing for spikes at round numbers (specifically, multiples of 10), and interpolate between 90 and 150 beds to obtain the counterfactual distribution in California in the absence of regulation. We obtain similar results for a range of values for $K = 1, \dots, 10$, although confidence intervals for the interpolated region in the counterfactual distribution tend to get large for $K \geq 10$. So, for brevity, we show results for $K = 5$ here. Appendix Figure C.1 shows that the counterfactual density of beds tends to be greater than the observed distribution above the threshold (other than the spike in the observed distribution around 120) before eventually converging with the observed distribution again.

Appendix Figure C.2 reproduces Figure 4, showing the share of bunchers at various observed and counterfactual sizes. Appendix Figure C.2a shows that below the threshold, the fraction of facilities that are bunchers increases as we approach the threshold, and that over 90% of facilities at the threshold of 99 beds are bunchers, echoing our main results using other states as the counterfactual and supporting the decision to treat 97–99 beds as the bunching region.

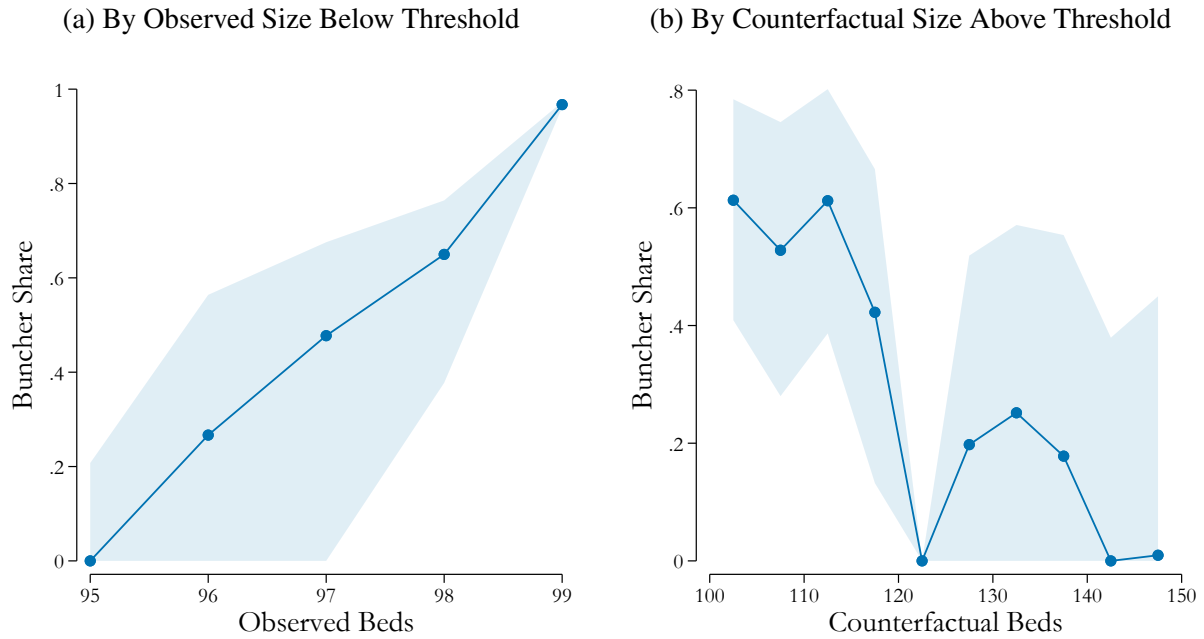
Appendix Figure C.2b reports the share of facilities that would counterfactually have bed counts above the threshold that instead bunch. As in our main results, we estimate that roughly 60% of facilities with preferred sizes of 100 to 119 beds are bunchers using these alternative methods. However, the alternative methods result in an estimate of no facilities bunching from 120–124 beds, while our main estimate is over 60%. This is because the traditional interpolation methods cannot account for the bunching at 120 beds corresponding to the federal social worker staffing requirement. This difference highlights the advantage we have in our setting of being able to rely on the distribution of facility sizes in other states, rather than having to rely on traditional interpolation methods. Nonetheless, these traditional methods also result in the estimated manipulation region ending around 135 to 140 beds.

Figure C.1. Observed and Counterfactual Distribution of Beds in California (Based on Polynomial Interpolation)



Notes: Figure plots observed (solid blue line) and counterfactual (orange dashed line) density of bed counts for California nursing homes using a fifth-order polynomial fit to facility-year counts at bed values outside the manipulation window [90, 150), with separate intercepts allowed at multiples of ten, and interpolated through the window. The sample is limited to facility-years for facilities in California with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019. 95% confidence intervals are from 2,000 bootstrap iterations at the facility level.

Figure C.2. Share of Facilities that are Bunchers



Notes: Panel (a) reports the estimated share of California nursing homes at each observed bed count that are bunchers, computed as the difference between the observed density in California and the counterfactual density implied by a polynomial interpolation, divided by the observed density following Equation (2). Shares are reported for 95–99 beds. Panel (b) reports the estimated share of California nursing homes at each counterfactual bed count that are bunchers, computed following Equation (3), in 5-bed bins. Shares are reported for 100–149 beds. The sample is limited to facility-years for facilities in California with non-missing size, facility characteristics, and ZIP characteristics for 2000–2019. 95% confidence intervals are from 2,000 bootstrap iterations at the facility level.